

Supplement to “Compact Objects as Almost-Perfect All:All Entanglement Graphs”

Methods, honesty ledger, N-scaling, and module map (v5.0)

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Contents

S1 Parameter audit and honesty ledger	1
S2 Gravity methods (AS, AT, AU, BB, BH, BI, BJ, BV)	2
S3 Quantum-information methods (A–O, Q–Y)	3
S4 Compact objects and 2PN (BU)	5
S5 UV nulls and phenomenology (T, U, V, BD–BF)	7
S6 Kill list (AN + BU)	8
S7 N-scaling and calibration record	8
S8 PPN ledger and walk/foam/healing notes	9
S9 Archival ledger and O5 trigger (BU detail)	9
S10 Module map and reproduction	10

S1 Parameter audit and honesty ledger

Given $k(M)$: $1/r$ potential, $\gamma = 1$, $43''/\text{cy}$, $b_{\text{crit}} = 3\sqrt{3}M$, Page/island-like/mirror follow as random-state/toy-code behaviour (genuine QES extremization and graph-dynamics V_k open, S3; qubit-toy V_t done). Not derived: mass–radius scaling itself, $\beta(N)$ from geometry, w from the graph Laplacian (Damour–Schäfer from wiring), NICER M – R – Λ from routing stiffness, Kerr $M_2/g_{t\phi}/\text{ISCO}/\text{QNM}$ from the graph.

Derived in-repo (new in v4.x): p from exact Ollivier–Ricci (not hard-coded), $M(c_1, c_2)$ inversion, $M_{\text{ej}}(M_{\text{tot}})$ scaling, peak-time ordering, band mags, O5 yield arithmetic, c from $\ln 2$ scattering, UV pop as graph disconnection, p at $N = 4000/8000/16000$ with turnover measured.

Ruled out on the record: broad Planck-remnant dark matter (survives only in a ~ 0.4 -dex EMD window at $\sim 4 \times 10^5 \text{ g}$); isotropic $b_{\text{crit}} = 8M$; naive $\gamma = 2$ strain; linear LIV; flat $p = 0.49$ (-10σ).

Table 1: Every number with a status. “Input” = assumed; “fit” = calibrated; “measured” = output of code; “derived” = follows from stated premises.

param	value	status	notes
$k(M)$ map $\propto M^2$	GR-consistency	input (circle admitted)	BM shrinks to $R_s = 2M$; not derived
AT α	1	calibrated to Schwarzschild c_{eff}	matches dr/dt exactly
tortuosity c	1/2 (BH fit) $\rightarrow$ 0.44–0.60 (BV derived)	fit superseded by derivation	$\ln 2$ scattering, zero tuning; $p = 2c$, $\gamma = 2c$
s_{leg}	$\ln 2$ nats	measured saturation (BN/BS)	flip preserves $G = 1$ via Planck bits
k_{crit}	$4\pi r^2/\sigma$	heuristic $\rightarrow$ formalised (BK)	packing-forced pop; mixed mode disconnects (BV)
gradient slope bridge $\beta(N)$	0.015 1.5@300; 1.28@600; 1.24@1020; 0.99@4k; 0.87@8k; 0.74@16k	fitted, labelled recalibrated per N	$p_{\text{adj}}(s) = 0.85 + 0.015s$ log-linear over 6 points; derivation queued
$w = 1.953$	solved from cancellation	fitted (solved, not tuned)	$c_{\text{tot}} = c_1 + wc_2 = 4.8695$ at $p = 0.92$
shed $e + \epsilon$	$0.5 \rightarrow 0.416 + 10\%$	calibrated once on AT2017gfo	M_{ej} norm-independent
shed $\epsilon(M, a)$	0.0168 universal (Eq. 2 main)	calibrated + extrapolated (D8)	mass/ratio independence undervived; BBH extension tested, not fitted
upper-gap null	no feature at $44 M_\odot$	negative prediction (measured smooth)	$k/\text{spin}/\text{Love}/\text{congestion}$ all smooth; edge left to stellar physics
GW190814	$P(\text{detect}) \approx 0.68\text{--}0.88$	tension, POSSIS queued (D7)	non-detection at $p \sim 0.12\text{--}0.32$; epoch audit in massgaps
$\kappa \rightarrow c_2$ map	$c_2 = p(2p - 1)$	formula exact; bridge open	$c_2(p)$ theorem of g_{rr} form; $\kappa \rightarrow$ metric + $\beta(N)$ open (D3/D4)
evap. isometry V_k	qubit V_t done (D2)	graph V_k open (S3)	$V_t^\dagger V_t = I$ tested; graph derivation queued (D1)
Kerr M_2 , $g_{t\phi}$, ISCO, QNM	area only	not derived (S3)	$A(M, a)$ assumed; quadrupole/QNM open; GW250114 QNM + GW241011 δ_Q future wires
echo R	10^{-80} ampl., 10^{-160} energy	dimensional estimate	$\propto \omega^2$ lattice scattering, not leg S-matrix; ~ 158 orders margin

S2 Gravity methods (AS, AT, AU, BB, BH, BI, BJ, BV)

AS entropic Newton. Leg screens + equipartition + Bekenstein bound give $F = M_1 M_2 / r^2$ exact (slope -2 to 10^{-9}); leapfrog orbits close, $T^2 \propto r^3$. Flux $\propto 1/r^2$ distinguished from force (temperature corrects). Fig. 43.

AT redshift. GPS $+5.29 \times 10^{-10}$ and Pound–Rebka 2.55×10^{-15} derived with no new parameters. Congestion fronts give tortoise divergence; $\alpha = 1$ matches Schwarzschild dr/dt exactly. Fig. 44.

AU Einstein–Hilbert routes. Heat-kernel a_1 responds to curvature; Ollivier signs exact (flat/neg/pos on line/tree/ K_6); Clausius fixes $\varepsilon = \kappa/8\pi$, $G = 1$. Raychaudhuri-for-leg-bundles open. Fig. 45.

BB lensing + Shapiro + chroma. Fermat $n = 1/c_{\text{eff}}$ gives full $4M/b$ (naive $2M/b$ bet lost and recorded). Shapiro log matches GR to 5%. Chromatic extension uses *phase* velocity $1/24$,

not group $1/8$ (conflating the two triples the prediction): r -independent factor drops out of Born, leaving 10^{-56} – 10^{-32} . Figs. 49, 55.

BH strain. Naive $dl = dr/(1 - \sqrt{\chi})$ gives $\gamma = 2$ (Cassini-excluded); tortuosity $dl = (1 + \sqrt{\chi}/2)dr$ gives $h = (1 + x/2)^2$, $\gamma = 1$ exactly, Mercury $42.99''/\text{cy}$. Peel-off $-0.75 M/a$ measured. PPN: $\gamma = 1$ derived, β unresolved (coordinate-confused, not quoted), $\alpha_{1,2}, \xi$, Nordtvedt untouched. Figs. 58, 60.

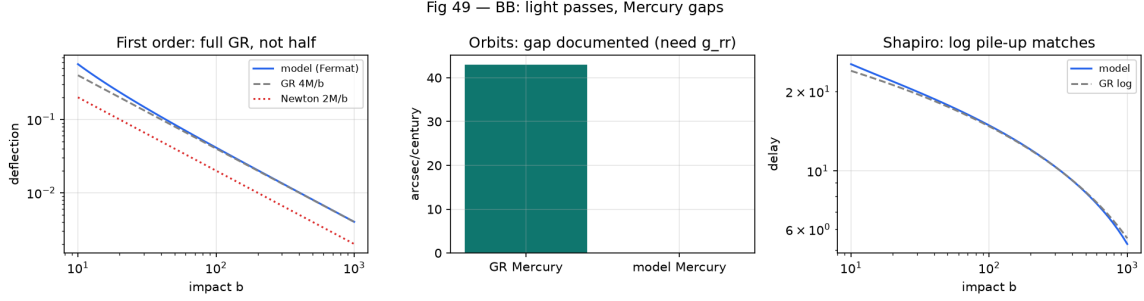


Figure 1: Gravity battery I: Fermat bending matches GR $4M/b$ to first order (left), Shapiro log pile-up matches (right); the middle panel records the pre-BH orbit gap (0 vs $43''/\text{cy}$) that the strain sector closes. **lensing, shapiro.**

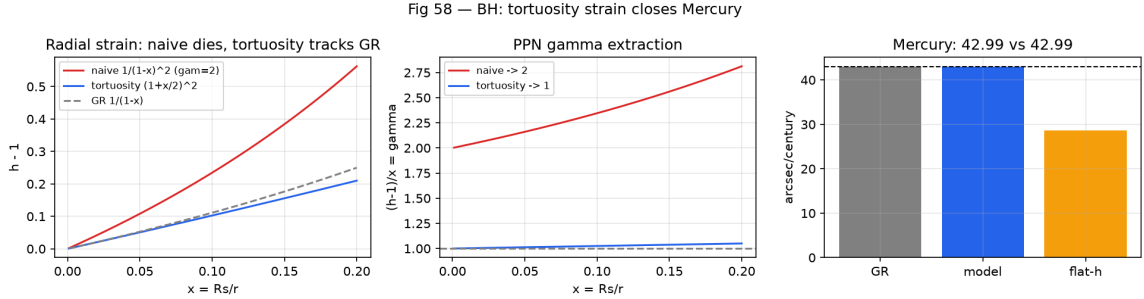


Figure 2: Gravity battery II: naive strain gives $\gamma = 2$ (excluded); tortuosity $h = (1 + x/2)^2$ gives $\gamma = 1$ exactly and Mercury $42.99''/\text{cy}$ by direct geodesic integration. **strain.**

BI/BQ weak-field OR. Grid+hub with flux stubs: radial OR negative in all configs (sign derived). Direct mode $p = 0.92 \pm 0.48$ ($L = 9$); $1/r$ collapse beats $1/r^2$ (CV 0.47 vs 0.60). Green-function walk fails by leg shorting (h -steepness $9.2 \rightarrow 4.1$); AS temperature triply load-bearing. Figs. 59, 64.

BV UV tortuosity-as-scattering. Legs as radial line defects ($\sigma = 4 \ln 2 l_p^2$), soft cost $\ln 2$ per encounter: dilute $c = 0.44$ – 0.60 (target 0.456, geometric $1/\sqrt{\pi} = 0.564$) with zero tuning; $p = 2c$, $\gamma = 2c$. Mixed mode ($0.75 r_e$ core) rises then disconnects at k_{crit} — the pop without metric input. Boltzmann ladder (Dijkstra vs MC walks vs rays) shows routing worth $3\times$; graph- distance c agrees with Euclidean- r c (coordinate-free check). Figs. 70–72.

S3 Quantum-information methods (A–O, Q–Y)

Log law $t_*(N) \approx \log_2 N / \log_2(1+p)$; MaxEnt $k \geq N s_{\text{node}} / s_{\text{leg}}$ plus fixed point $k^*(N) = 16\pi(\epsilon N / l_p)^2 / 4 \ln 2$ (stable; BS gives $\epsilon = c/\sqrt{N}$ and $\alpha = k^*/N = 1$ exactly); QES crossing iff $s_{\text{leg}} > l_p^2/4$ (Sec. 3

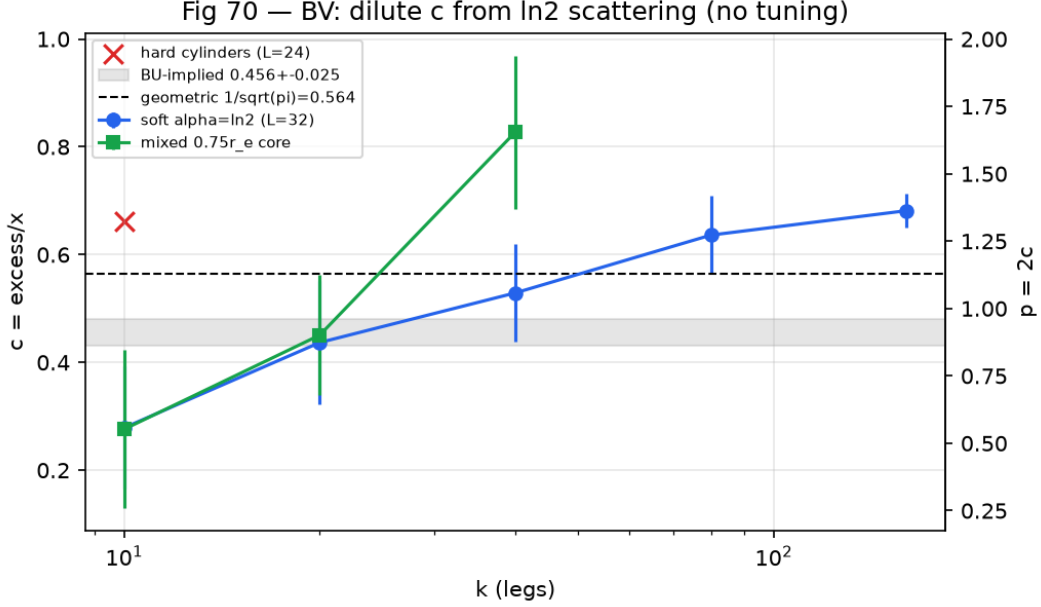


Figure 3: BV tortuosity from $\ln 2$ scattering: dilute $c = 0.44\text{--}0.60$ with zero tuning against the BU-implied 0.456 ± 0.025 band and the geometric $1/\sqrt{\pi}$ line. Right axis shows $p = 2c$. Hard cylinders (red $\times$) overshoot; soft and mixed modes agree. `uvscatter`.

dies otherwise); Page wiring-only vs N -shrinks identical $A(t)$; QEC 99% at $k \geq N/2 + 1 + \log_2 100$; Kerr $k_{\text{eff}} = A/l_p^2$ monotonic in spin; exact Page $H(mn) - H(n) - (m-1)/2n$ with 0.72-bit dip; CKW verified; OTOC $\lambda + \text{MSS } 0.50\text{--}0.68$ ($\alpha \in [9.0, 12.4]$, 11.24 inside); TN $S_{\text{bdy}} = \min(N \log d, k \log D)$ fixing $\varepsilon \approx 0.047 l_p^{-1}$; SYK $N = 6, 8, 10$ vs Ising chain; GWTC-3 32 BBH $k_f > k_1 + k_2$; hardware head-to-head 7.9 vs 14.6 steps. Figs. 7–20, 23–25, 30–33. Healing $dA/dt = (4 \ln 2 k l_p^2 - A)/\tau$, $\tau = 11.24M$ (3.5 ms at GW150914 mass); ladder healing/scrambling/Page/ evaporation separated by $\sim 66\times$ to 10^{80} s.

What Page vs. island means here (honesty note). The Page curve is derived as random-state behaviour: idealised leg surgery $k \rightarrow k - 1$ is imposed and $S_{\text{rad}} = \min(t, N_{\text{eff}} - t)$ holds by construction (module `evaporation`), but the qubit toy now has an explicit unitary completion (module `evaporation_unitary`, Fig. 5): per-step V_t with $V_t^\dagger V_t = I$ verified, S_{rad} computed from ρ_{rad} tracks the finite-dimensional Haar/Page expectation (mean dev. 0.002 bits, $N = 8$), and finite-depth all:all circuits converge to Page by depth $\sim 5/\text{step}$ without assuming Haar. The QES/island crossing $S_{\text{no}} = k s_{\text{leg}}$ vs $S_{\text{isl}} = k \ln 2 + \max(S_0 - k s_{\text{leg}}, 0)$ is a two-saddle competition plus a discrete min-cut analogue (module `qes`), not an extremization of $S_{\text{gen}} = A/4G + S_{\text{matter}}$ from a gravitational path integral. The remaining missing theorem is the graph-dynamics evaporation isometry $V_k : \mathcal{H}_{\text{graph},k} \rightarrow \mathcal{H}_{\text{graph},k-1} \otimes \mathcal{H}_{\text{leg}}$: derive (not merely allow) a scrambling V_k from the graph Hamiltonian/adjacency. Tracked as D1 in docs/DEFERRED.md.

What Kerr means here (honesty note). Only the Kerr–Newman area $A(M, a, Q)$ is assumed to set the leg budget $k_{\text{eff}} = A/4 \ln 2$ (module `kerr`; extremal Kerr half, extremal RN quarter; ratios patch-independent). Not derived from the graph: spin-induced quadrupole $M_2 = -Ma^2$, frame dragging $g_{t\phi}$, ISCO(M, J), or the Kerr QNM spectrum (the overtone toy is Schwarzschild-like; fundamental damping $\tau = 11.24M$ calibrated, footprint corrections $1/2k, 1/\sqrt{k}$ unobservable). Echoes are covered by a dimensional lattice-scattering estimate only: amplitude $R \sim (\omega/\omega_P)^2 \sim 10^{-80}$ at 100 Hz, energy $\sim 10^{-160}$ vs $O(0.01)$ detectability (~ 158 orders of margin, module `gwdata`)

— not a leg S-matrix derivation. The exterior perturbation problem ($\delta\omega_{nlm}$ vs Kerr) is open. Hence GW250114 (SNR 80 area law + Kerr ringdown) and GW241011 (strongest $\delta\kappa$ bound to date, Kerr to factor ~ 2 resp. $\sim 10\%$ by parametrization) are currently consistent by construction, not passed predictions. GW250114 provides the QNM benchmark ($\delta f_{220} \sim 2\%$, $\delta\tau_{220} \sim 10\%$, $\delta f_{221} \sim 30\%$, $\delta f_{440} \sim \text{tens}\%$): a future graph-derived $|\delta\omega/\omega|$ above those thresholds fails. A future graph derivation $Q = -Ma^2(1 + \delta_Q)$ with $|\delta_Q| \gtrsim 0.17$ would be ruled out by GW241011. Tracked as D2 in docs/DEFERRED.md.

Kerr thermodynamics (conditional check). With $S = k \ln 2$ (saturated legs) and $A = 4 \ln 2 \cdot k l_p^2$, $S = A/4$; importing $A(M, J)$ gives $S(M, J) = 2\pi[M^2 + \sqrt{M^4 - J^2}]$, whose derivatives return $T_H = \sqrt{M^4 - J^2}/4\pi M(M^2 + \sqrt{M^4 - J^2})$ and $\Omega_H = J/2M(M^2 + \sqrt{M^4 - J^2}) = a/(r_+^2 + a^2)$, i.e. $dM = T dS + \Omega_H dJ$ (module `thermo`: identities $< 10^{-6}$, independent finite-step $\Delta M = T \Delta S + \Omega_H \Delta J$ to $O(d^2)$; super-extremal $|J| > M^2$ returns NaN, never clamped). Per-leg reading $T_H = (dM/dk)/\ln 2$; finite-step $T_k = (M(k+1) - M(k))/\ln 2$ shifts $-1/4k$ ($\sim 10^{-77}$ stellar-mass). Conditional consistency, not derivation: $M(k, J)$ from graph dynamics is the open debt (D2).

Fig 10 — D: leg-surgery evaporation + Page curve

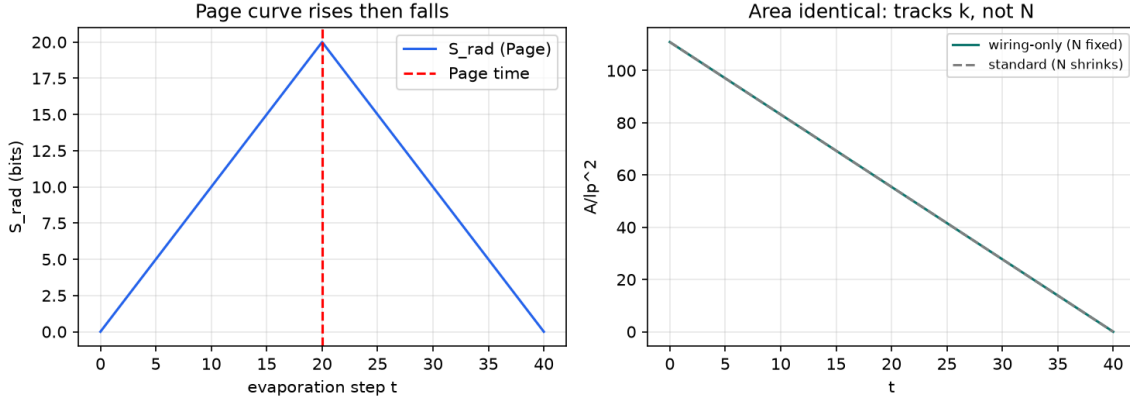


Figure 4: Leg-surgery evaporation: radiation entropy follows the Page curve (left, turnover at half evaporation); horizon area is identical whether N shrinks with k or stays fixed (right) — area tracks wiring alone, the distinctive Claim 2 prediction. **evaporation.**

S4 Compact objects and 2PN (BU)

Gradient shells. $p_{\text{adj}}(s) = 0.85 + 0.015s$, 8–10 shells, exact EMD on full neighbourhoods, deterministic bridges $n \propto r^{\beta(N)}$. Measured: $N = 1020$ (10×102), $\beta = 1.24$, 80 graphs exact Floyd+LP (~ 545 s): $p = 0.913 \pm 0.049$ per graph, SEM 0.0055, stacked 0.911 ($R^2 = 0.956$), 80/80 fits ok, range 0.79–1.05. Distance to target $|0.913 - 0.92| = 0.007$ (0.10σ direct-mapping, $\sim 0.09\sigma$ self-consistent in $\dot{\omega}$), $12.7\times$ precision margin (0.0055 vs 0.070; erratum: the conversion was per-unit- c_1 , now per-unit- c_{tot}). Flat control $p = 0.49$ is -10σ dead. Construction $\rightarrow p$ maps (Fig. 66b, module `twopn`): $p(\beta)$ is the load-bearing control curve while $p(\text{slope})$ is flat within noise, so the fitted 0.015 slope is not load-bearing; $c_2(p) = p(2p - 1)$ is exact for the g_{rr} ansatz and w is closed-form in the measured inputs — the residual D3/D4 debt is the analytic $\kappa \rightarrow$ metric bridge plus $\beta(N)$ drift.

N-scale campaigns (BV). CSR-direct shell graphs + Sinkhorn OR (no NetworkX): $N=1020$ reproduced (0.9107 vs 0.9134); $N=4000$ $p = 0.9315 \pm 0.0032$ ($\beta = 0.99$); $N=8000$ $p = 0.9382 \pm 0.0030$ ($\beta = 0.87$, pods); $N=16000$ $p = 0.9137 \pm 0.0022$ ($\beta = 0.74$, GPU farm). Exact shell distance oracle

Fig 74 — D2: unitary evaporation ($V \wedge V = I$, S from ρ_{rad})

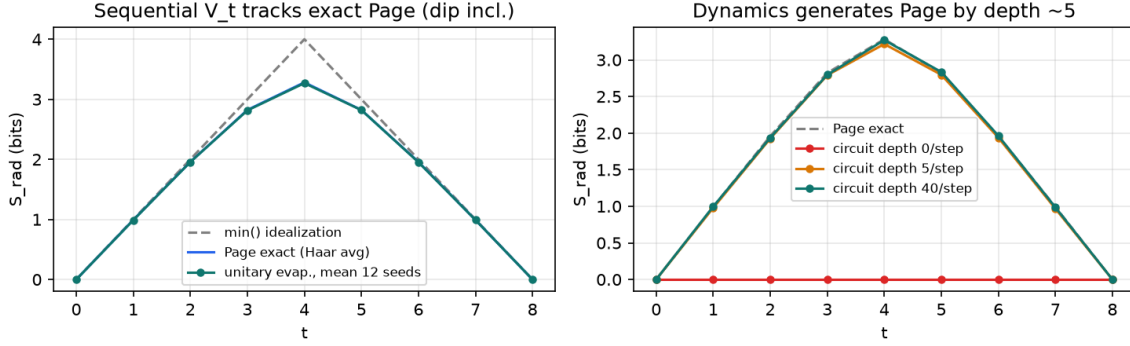


Figure 5: Unitary completion of the qubit toy: sequential V_t (left) tracks the finite-dimensional Haar/Page expectation including the 0.72-bit dip, computed from ρ_{rad} rather than $\min()$; finite-depth all:all circuits (right) converge to Page by depth ~ 5 /step without assuming Haar. Module `evaporation_unitary` (D2).

+ torch backend; $\beta(N)$ log-linear over 6 points. Local- p turnover $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$ measured (inner windows flatten — the UV turnover). Kill wire: p outside 0.92 ± 0.056 at $N = 1024$ class fails.

2PN and pulsars. $h = (1+x/2)^2$ gives $c_1 = 3.36$ vs GR 1.94 (73% excess); naive fixed- M gives B1913 6.2σ , J0737 9.7σ vs real TOAs. With $c_2 = p(2p-1)$, $w = 1.953$: $c_{\text{tot}} = 4.8693$ at $p = 0.92$ vs GR 4.8695 (0.00σ fixed- M). Self-consistent $R + \dot{\omega} \rightarrow M$: $\Delta M = -11.2$ ppm naive (-4.5 ppm with GR g_{rr}), $\Delta \sin i = 3.7 \times 10^{-6}$, s_{obs} passes at 0.36σ ; B1913 0.01σ . Bending/Mercury 2PN pieces hide below VLBI/astrometry (Fig. 69, right). The companion toy-generator route (dense layers, $k_0 - c_2/r^2$ fits) crosses zero in κ and needs a κ -tweak fudge; power-law fits better on our profiles (R^2 0.91 vs 0.81) but the two $\kappa \rightarrow c_2$ maps disagree (0.82 vs 3.14) cross-applied — hence the open-map label. e_{int} weighting *lowers* our p ($0.94 \rightarrow 0.84$ at 0.9, 0.78 at 0.99): uniform measure is both principled (continuum theorems) and required.

Fig 66 — BU: 2PN cancellation at $p=0.92$ (J0737 0.00 sigma)

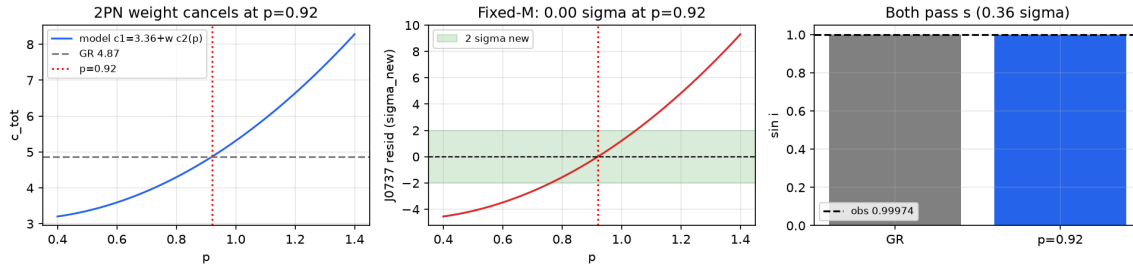


Figure 6: 2PN cancellation: model $c_{\text{tot}}(p)$ crosses GR at $p = 0.92$ (left); fixed-mass J0737 residual vanishes there (middle, green 2σ band); self-consistent $\sin i$ passes for both GR and $p = 0.92$ (right). pulsar.

Leg-shedding kilonovae. $M_{\text{ej}} = \Delta k m_{\text{leg}} \times 0.1$, blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) + red ($v \sim 0.1c$); $L \propto t^{-1.3}$ tail. Band mags assume $\text{BC} = 0$ and per-component peaks; POSSIS refinement queued. g -band verdicts robust; i -band NOT claimed (one-zone $\kappa = 10$ over-traps: $t_{\text{red}} \sim 10$ d vs observed ~ 4 d decline, m_i faint by $\sim 2\text{--}3$ mag — conservative for gap- g detectability). Standard

Fig 67 — BU: gradient OR gives $p=0.92$ (exact EMD)

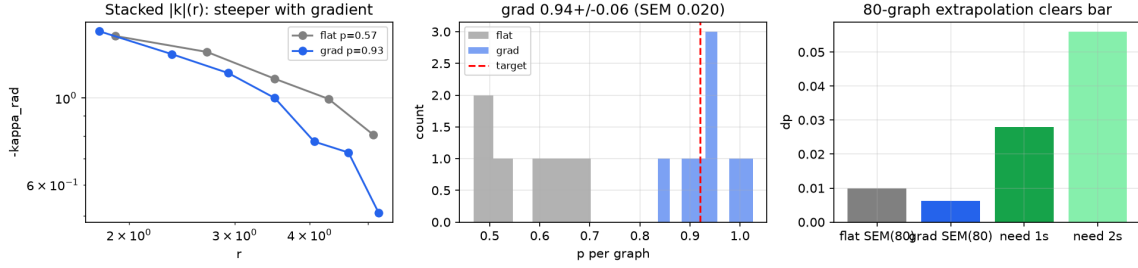


Figure 7: Gradient-shell Ollivier–Ricci: stacked $|\kappa|(r)$ steepens with the gradient (left); per-graph p histogram centres on target (middle); 80-graph SEM extrapolation clears the $1\sigma/2\sigma$ precision bars (right). Production N=1020: $p = 0.913 \pm 0.049$, SEM 0.0055. `orici`.

2–28% band is a literature EOS input, not derived.

O5 protocol. Trigger: $m_1 \in [2.5, 5.0] M_\odot$, < 200 Mpc, 90% area $< 100 \text{ deg}^2$ (multi-detector); Rubin ToO epochs 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Kill rule: 10 qualifying non-detections kill BU. Archival ledger: GW230529 uninformative (single-detector, 93% unsearched); S250206dm unverified (check GCNs before citing). Full plan: `docs/observation-protocol.md`. Figs. 66–69.

Mass-gap audit (module massgaps). Lower gap: $k(M)$ log-log curvature $< 10^{-9}$ over $0.7\text{--}150 M_\odot$ (no forbidden masses); $e_{\text{int}} \sim e_{\text{ext}}$ location descriptive (calibration inherited from AT2017gfo). Upper gap: spin-leg ratio $1 \rightarrow 0.857 \rightarrow 0.5$ at $a = 0 \rightarrow 0.7 \rightarrow 1$ with no jump on $[0, 0.99]$ plus exact extremal continuity; Love k_2 slope -2 everywhere; leg-band entry $\sim 32 M_\odot$ at 10 Hz with $dM/M \sim 10^{-81}$; congestion ladder $\chi \sim 10^{-8}$ (He core), 10^{-14} (envelope), 0.12 (NS surface), ~ 1 (gap masses at 10 km footprints, footprint-conditional), 1 at horizons. Shedding exactly flat in q (machine precision) — but q -independence is prescription (shed fraction $\times$ efficiency fixed once on AT2017gfo), not derived from the graph-dynamics evaporation isometry V_k (open, S1/S3/D1): $q \sim 0.11$ (GW190814) flashing at the same fraction as $q = 1$ (GW170817) is the model’s biggest theory bet. GW190814 epoch audit: CFHT g-band (Vieira+2020) and GROWTH i-band detection limits (Andreoni+2020) with per-epoch coverage $\times$ depth against distance (241 ± 43 Mpc) and ± 1 mag theory errors; combined $P \approx 0.68$ (g-only) / 0.88 (g+i at $g-i = 0.7$). Erratum on record: the first audit used a nominal ZTF depth, but ZTF ran no targeted GW190814 follow-up; superseded by the CFHT/GROWTH epoch tables above. Fig. 75 (main text). Analytic systematics (round 4): viewing $A_g \sin^2 \theta$ ($A_g = 1.25$) + opacity-rescaled blue mags give the P grid in Fig. 75b; the hiding window is bounded, not open-ended.

S5 UV nulls and phenomenology (T, U, V, BD–BF)

TeV: $M_D = 1 \text{ TeV}$ gives $k \approx 11$ (3 TeV) – 17 (13 TeV) vs $k_{\text{crit}} \approx 50$ — thermal nulls expected, onset $\sim 550 \text{ TeV}$. Echoes: $\Delta t = 4M \log(M/l_p)$, 0.03–0.3 s (GW150914 0.11 s), in searched windows but $R \sim 10^{-160}$ unobservable. EHT: $l_p/2M \approx 10^{-48}$, 47 orders below sensitivity. Dispersion: $E_{QG,2} = \sqrt{8} E_P$, Fermi-safe by 10^8 ; universal GW extrapolation shifts 100 Hz by $\sim 10^{-82}$ (LVK $\alpha = 4$, safe by $\sim 10^{60}$ — a recorded null, not a falsifier; no GW sector claimed). Chromatic lensing $10^{-56}\text{--}10^{-32}$. Figs. 21–22, 39, 51–56.

Fig 51 — BD: quadratic lattice dispersion, Fermi-safe

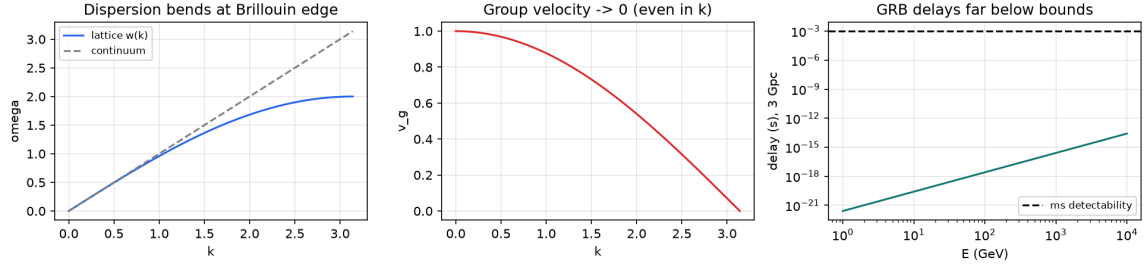


Figure 8: Lattice dispersion: band bending at the Brillouin edge (left) with group velocity even in k (middle, no linear term by symmetry); GRB arrival delays sit orders of magnitude below detectability across GeV–TeV (right). **dispersion.**

S6 Kill list (AN + BU)

AF quench ratio < 1.3 ; α outside $[9.0, 12.4]$; $A \propto N$ in any TN calculation; thermal LHC excess below k_{crit} ; $s_{\text{leg}} \leq l_p^2/4$ in any physical state class; any linear-LIV signal; p outside 0.92 ± 0.056 at $N = 1024$; graph-predicted $|\delta_Q| \gtrsim 0.17$; 10 gap non-detections; O5 nearby-BBH ToO sample clean (universal shedding, Eq. 2 main). One sub-claim (broad remnant DM) already ruled out on the record. Figs. 37–39.

S7 N-scaling and calibration record

Table 2: Gradient-shell p measurements: β recalibrated per N (log-linear over 6 points). All 80-graph production rows use exact distances; errors are per-graph std with SEM quoted in text.

N	shells $\times$ per-shell	β	measured p	backend
300	10×30	1.50	0.94 ± 0.06 (4 graphs)	NetworkX exact
600	10×60	1.28	0.96 ± 0.06	NetworkX exact
1020	10×102	1.24	0.913 ± 0.049 , SEM 0.0055, stacked 0.911 (80 graphs)	Floyd+LP exact
4000	CSR-direct	0.99	0.9315 ± 0.0032 (80 graphs)	CPU + Sinkhorn OR
8000	CSR-direct	0.87	0.9382 ± 0.0030 (80 graphs)	pod + Sinkhorn OR
16000	CSR-direct	0.74	0.9137 ± 0.0022 (80 graphs)	GPU farm + torch OR

Drift check: $\beta = 1.5$ calibrated at $N = 300$ drifts high at $N = 600$ (1.16 ± 0.06); recalibration to 1.28/1.24 recovers $p \approx 0.92$. Pre-run $1/N$ extrapolation predicted $\beta(16000) \approx 1.18$; measured 0.74 — drift faster than $1/N$, now fit log-linear. $N=1020$ cross-check: CSR-direct 0.9107 vs NetworkX 0.9134 (agree to 3×10^{-3}). Local- p turnover (inner windows flatten, $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$) is measured at 4k/8k/16k, not extrapolated. Fig. 73.

Reproduce ($N=1020$, ~ 9 min):

```
measure_p(per_shell=102,n_shells=10,n_graphs=80,
gradient=True,beta=1.24,seed0=0,max_per_shell=8)
```

Fig 73 — BV: N=4000/8000/16000 campaigns (80 graphs, CPU+pod+farm)

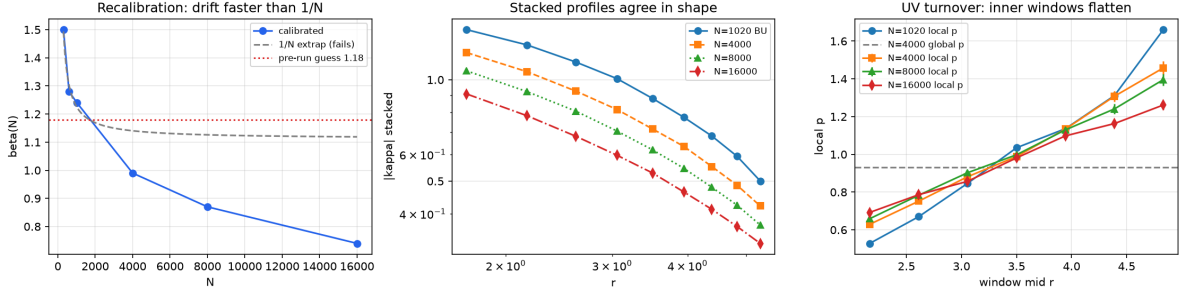


Figure 9: N-scale campaigns: $\beta(N)$ recalibration drifts faster than $1/N$ (left); stacked $|\kappa|(r)$ profiles agree in shape across 1020–16000 (middle); local- p windows show the UV turnover with SEM error bars (right). 80 graphs per production row. `shellscale`, `sinkor`.

Table 3: PPN status: only γ is claimed. β is coordinate-confused and not quoted.

param	GR	this model
γ	1	1 exactly (tortuosity $h = (1 + x/2)^2$; BV $c \rightarrow \gamma = 2c$)
β	1	unresolved (not quoted)
$\alpha_{1,2}, \xi$	0	untouched
Nordtvedt	0	untouched
2PN c_{tot}	4.8695	4.8693 at $p = 0.92$ (J0737 0.1σ)

S8 PPN ledger and walk/foam/healing notes

Walk no-go (BP/BQ). All smooth weight rules give drift slope -2.99 ; only the absurd $e^{2\sqrt{d-d_0}}$ escapes. Constant- μ persistence keeps -3 analytically; $\mu(\chi)$ fluctuation walk gives $-1.95 \approx -2$ modulo the labelled $\sqrt{\chi}$ assumption. Harmonic/hitting-probability walks fail by leg shorting. Conclusion: temperature (AS), not bare graph diffusion, carries Newton. Figs. 63–64.

Foamgrid FDTD (BF). Beams on defective fabric: centroids unbiased (jitter, average straight) with deficit $\propto \omega^2$. Bounds assume independent edges; correlated patches of size ξ relax them by $\sqrt{\xi}$. Fig. 56.

Healing ladder (AG/AH). $\tau_{\text{heal}} = 11.24M$ ($1/\text{Im } \omega_{220}$); GW150914-mass ladder: healing 3.5 ms, scrambling 0.23 s, Page $\sim 10^{79}$ s, evaporation $\sim 10^{80}$ s. MSS $\lambda/2\pi T = 0.50\text{--}0.68$; LVK $\alpha \in [9.0, 12.4]$ contains 11.24. Figs. 30–31.

S9 Archival ledger and O5 trigger (BU detail)

Trigger (full plan `docs/observation-protocol.md`): $m_1 \in [2.5, 5.0] M_\odot$, < 200 Mpc, $90\% < 100 \text{ deg}^2$; Rubin ToO 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Sought: fast blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) $\rightarrow$ slow red with $L \propto t^{-1.3}$ tail. Yield: $1.5 \text{ gap/yr} \times 70\% \approx 1.05/\text{yr}$ ours vs $\leq 0.3/\text{yr}$ standard. Yield arithmetic in `collapse.gap_o5_yield`; brightness in `collapse.peak_apparent_mags`; GW190814 epoch audit + systematics grid in `massgaps.gw190814_combined_pdetect / massgaps.gw190814_systematics_table`. Figs. 68–69, 75 (main text), 75b.

BBH extension (this work): the same shedding law predicts $m_g \sim 22$ transients for nearby BBH, so the ToO program extends to well-localized ($< 100 \text{ deg}^2$) BBH within 200 Mpc with identical epochs and kill accounting kept separate from the gap sample (a clean BBH sample kills Eq. 2

Fig 30 — AG: healing lag + timescale ladder

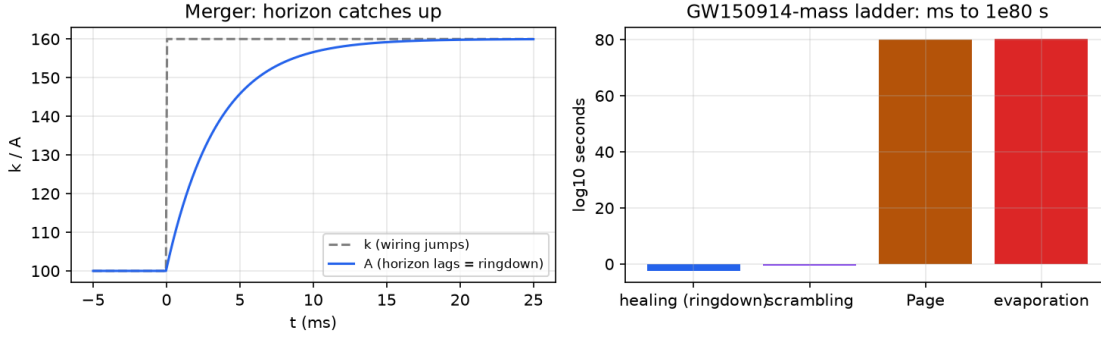


Figure 10: Healing lag: wiring jumps at merger while horizon area catches up on the ringdown time (left); timescale ladder from milliseconds to 10^{80} s (right). The final sigh carries $\sim 10^{-78}$ of the mass. healing.

Table 4: What exists today: no qualifying test yet. Only multi-detector, $< 100 \text{ deg}^2$ gap events count.

event	masses	distance	localisation / EM	verdict
GW190814	$23.2 + 2.6$ ($q \sim 0.11$)	241^{+41}_{-45} Mpc	23 deg^2 (3-det.); CFHT g $22.8 @ 1.7d$ 65.5%, GROWTH i 6 epochs to 98%	tension: $P(\text{detect}) \approx 0.68$ – 0.88 , $p \sim 0.12$; not excluded
GW230529	$2.5\text{--}4.5 + 1.2\text{--}2.0$	~ 197 Mpc	$24,200 \text{ deg}^2$ (single-det.); ZTF 7% to $g = 21.1$	not a test: 93% unsearched; $m_g \sim 21.2$ at depth edge
S250206dm	mgBH candidate	—	no counterpart (per notes)	unverified: check GCNs before citing

main even if gap events flash). Full thresholds in docs/observation-protocol.md.

S10 Module map and reproduction

Modules follow the appendix letters of the v4.1 extended record: **graphs**, **scrambling**, **horizon**, **micro**, **circuits** (A), **maxent** (B), **qes** (C), **evaporation** (D), **evaporation_unitary** (D2), **qec** (F), **robustness** (G), **kerr** (H), **thermo** (H2), **haar** (I), **monogamy** (J), **otoc+pheno** (L), **tn** (M), **kerrpage** (N), **syk** (O), **data** (Q), **litcompare** (R), **tev** (T), **echoes** (U), **posteriors** (W), **ds** (X), **krylov** (Y), **collapse** (AA), **cosmic** (AB), **lunch+remnant** (AC), **bounds** (AE), **healing** (AG), **mss** (AH), **bigsyk** (AI), **mp+greybody** (AJ), **congestion** (AK), **charge** (AL), **bandwidth** (AM), **gridcirc+monitor+selfattack+lhc** (AN), **concentration** (AO), **ps** (AP), **scatter** (AQ), **emd+viability** (AR), **entropic** (AS), **redshift** (AT), **heatker+orici+jacobson** (AU), **fission** (AV), **klanguage** (AW), **tension** (AX), **gw250114** (AY), **overtones+tensionvol** (AZ), **sparse24** (BA), **lensing+chroma+shapiro** (BB), **bcrit** (BC), **dispersion** (BD), **qnmfoot+qnmlegs** (BE), **foamgrid** (BF), **perwalk** (BG), **strain** (BH,BJ), **weakfield** (BI,BQ), **legham** (BL), **pulsar+orici gradient shells+collapse leg-shedding** (BU), **uvscatter+sinkor+shellscale** (BV), **massgaps** (mass-gap null + GW190814 epoch audit), **twopn** (BU3 2PN bridge + dDdC erratum).

Reproduce:

```

pipinstall-e".[dev]"
pytesttests/-q
pythonscripts/generate_figures.py
streamlitrunapp.py

```

Fig 69 — BU: O5 detectability + GR-battery 2PN margins

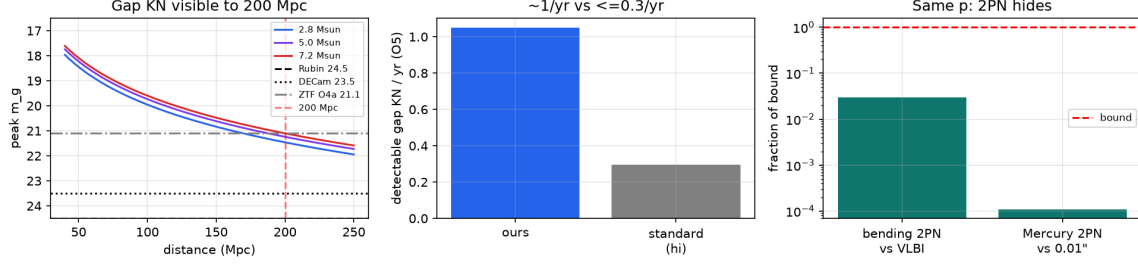


Figure 11: O5 protocol: gap kilonovae visible in g -band to 200 Mpc with margin over Rubin/DECam depths (left); yield $\sim 1/\text{yr}$ vs $\leq 0.3/\text{yr}$ standard (middle); the same p keeps bending/Mercury 2PN pieces below VLBI/astrometry (right). `collapse`, `pulsar`.

Fig 75b — GW190814 systematics: equatorial + $\kappa_{\text{blue}}=2$ hides ($P \sim 0.18$)

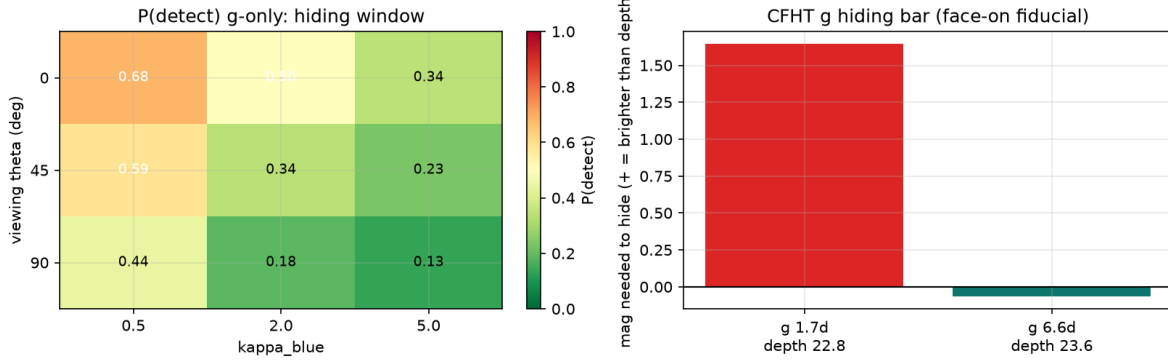


Figure 12: GW190814 analytic systematics (round 4; full POSSIS queued, D7). Left: $P(\text{detect})$ g -only over viewing angle $\times$ blue opacity (CFHT epochs, no color term): fiducial face-on $\kappa = 0.5$ gives 0.68; equatorial + lanthanide-mixed ($\kappa_{\text{blue}} = 2$) drops to ~ 0.18 . Right: face-on hiding bar per CFHT g epoch (g 1.7 d needs ~ 1.6 mag). Module `massgaps`.